■ 3.2.9 Orthogonal Polynomials

`LegendreP[n, x]`	Legendre polynomials $P_n(x)$
`LegendreP[n, m, x]`	associated Legendre polynomials $P_n^m(x)$
`SphericalHarmonicY[l, m, theta, phi]`	spherical harmonics $Y_l^m(\theta, \phi)$
`GegenbauerC[n, m, x]`	Gegenbauer polynomials $C_n^m(x)$
`ChebyshevT[n, x]`, `ChebyshevU[n, x]`	Chebyshev polynomials $T_n(x)$ and $U_n(x)$ of the first and second kinds
`HermiteH[n, x]`	Hermite polynomials $H_n(x)$
`LaguerreL[n, x]`	Laguerre polynomials $L_n(x)$
`LaguerreL[n, a, x]`	generalized Laguerre polynomials $L_n^a(x)$
`JacobiP[n, a, b, x]`	Jacobi polynomials $P_n^{(a,b)}(x)$

Orthogonal polynomials.

Legendre polynomials `LegendreP[n, x]` arise in studies of systems with three-dimensional spherical symmetry. They satisfy the differential equation $(1 - x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n + 1)y = 0$, and the orthogonality relation $\int_{-1}^{1} P_m(x)P_n(x)\,dx = 0$ for $m \neq n$.

The **associated Legendre polynomials** `LegendreP[n, m, x]` are obtained from derivatives of the Legendre polynomials according to $P_n^m(x) = (-1)^m(1 - x^2)^{m/2}\frac{d^m}{dx^m}P_n(x)$. Notice that for odd integers $m \leq n$, the $P_n^m(x)$ contain powers of $\sqrt{1 - x^2}$, and are therefore not strictly polynomials. The $P_n^m(x)$ reduce to $P_n(x)$ when $m = 0$.

The **spherical harmonics** `SphericalHarmonicY[l, m, theta, phi]` are related to associated Legendre polynomials. They satisfy the orthogonality relation $\int Y_l^m(\theta, \phi)Y_{l'}^{m'}(\theta, \phi)\,d\Omega = 0$ for $l \neq l'$ or $m \neq m'$, where $d\Omega$ represents integration over the surface of the unit sphere.

This gives the algebraic form of the Legendre polynomial $P_6(x)$.

```
In[1]:= LegendreP[6, x]

                                 2          4          6
              5         105 x      315 x      231 x
Out[1]= -(--) + ------- - ------- + -------
             16           16         16         16
```

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The integral $\int_{-1}^{1} P_5(x)\, P_6(x)\, dx$ gives zero by virtue of the orthogonality of the Legendre polynomials.

```
In[2]:= Integrate[LegendreP[5,x] LegendreP[6,x],
                                {x, -1, 1}]

Out[2]= 0
```

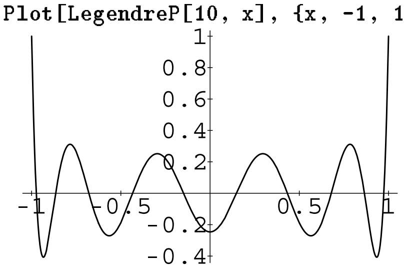
Integrating the square of a single Legendre polynomial gives a nonzero result.

```
In[3]:= Integrate[LegendreP[6, x]^2, {x, -1, 1}]

        2
Out[3]= --
        13
```

High degree Legendre polynomials oscillate rapidly.

```
In[4]:= Plot[LegendreP[10, x], {x, -1, 1}]
```



The associated Legendre "polynomials" contain pieces proportional to $\sqrt{1-x^2}$.

```
In[5]:= LegendreP[6, 3, x]

                       3
                       -           3
                   2   2  -945 x   3465 x
Out[5]= -((1 - x )    (------- + -------))
                             2         2
```

Section 3.2.10 discusses the generalization of Legendre polynomials to Legendre functions, which can have non-integer degrees.

```
In[6]:= LegendreP[6.1, 0]

Out[6]= -0.306319
```

Gegenbauer polynomials GegenbauerC[n, m, x] can be viewed as generalizations of the Legendre polynomials to systems with $(m+2)$-dimensional spherical symmetry. They are sometimes known as **ultraspherical polynomials**.

Series of Chebyshev polynomials are often used in making numerical approximations to functions. The **Chebyshev polynomials of the first kind**, ChebyshevT[n, x], are defined by $T_n(\cos\theta) = \cos(n\theta)$. They are normalized so that $T_n(1) = 1$. They satisfy the orthogonality relation $\int_{-1}^{1} T_m(x) T_n(x) (1-x^2)^{-\frac{1}{2}}\, dx = 0$ for $m \neq n$. The $T_n(x)$ also satisfy an orthogonality relation under summation at discrete points in x corresponding to the roots of $T_n(x)$.

The **Chebyshev polynomials of the second kind** ChebyshevU[n, z] are defined by $U_n(x) = \frac{\sin(n+1)\theta}{\sin\theta}$. With this definition, $U_n(1) = n+1$. The U_n satisfy the orthogonality relation $\int_{-1}^{1} U_m(x) U_n(x) (1-x^2)^{\frac{1}{2}}\, dx = 0$ for $m \neq n$.

The name "Chebyshev" is a transliteration from the Cyrillic alphabet; several

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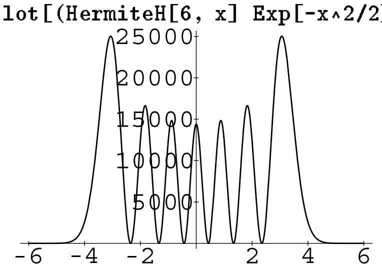
other English spellings, such as "Tschebyscheff", are sometimes used.

Hermite polynomials `HermiteH[n, x]` arise as the quantum mechanical wave functions for a harmonic oscillator. They satisfy the differential equation $\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2ny = 0$, and the orthogonality relation $\int_{-\infty}^{\infty} H_m(x)H_n(x)e^{-x^2}\,dx = 0$ for $m \neq n$. An alternative form of Hermite polynomials sometimes used is $He_n(x) = H_n(\frac{x}{\sqrt{2}})$ (a different overall normalization of the $He_n(x)$ is sometimes used).

The Hermite polynomials are related to the **parabolic cylinder functions** or **Weber functions** $D_n(x)$ by $D_n(x) = e^{-x^2/4}H_n(\frac{x}{\sqrt{2}})$.

This gives the probability density for an excited state of a quantum mechanical harmonic oscillator. The average of the wiggles is roughly the classical physics result.

```
In[7]:= Plot[(HermiteH[6, x] Exp[-x^2/2])^2, {x, -6, 6}]
```



Generalized Laguerre polynomials `LaguerreL[n, a, x]` are related to hydrogen atom wave functions in quantum mechanics. They satisfy the differential equation $x\frac{d^2y}{dx^2} + (a + 1 - x)\frac{dy}{dx} + ny = 0$, and the orthogonality relation $\int_{-\infty}^{\infty} L_m^a(x)L_n^a(x)x^a e^{-x}\,dx = 0$ for $m \neq n$. The **Laguerre polynomials** `LaguerreL[n, x]` correspond to the special case $a = 0$.

Jacobi polynomials `JacobiP[n, a, b, x]` occur in studies of the rotation group, particularly in quantum mechanics. They satisfy the orthogonality relation $\int_{-1}^{1} P_m^{(a,b)}(x)P_n(a,b)(x)(1-x)^a(1+x)^b\,dx = 0$ for $m \neq n$. Legendre, Gegenbauer and Chebyshev polynomials can all be viewed as special cases of Jacobi polynomials. The Jacobi polynomials are sometimes given in the alternative form $G_n(p,q,x) = \frac{n!\Gamma(n+p)}{\Gamma(2n+p)}P_n^{p-q,q-1}(2x-1)$.

You can get formulae for generalized Laguerre polynomials with arbitrary values of a.

```
In[8]:= LaguerreL[2, a, x]

          2
         x                   (1 + a) (2 + a)
Out[8]= -- - x (2 + a) + ---------------
         2                        2
```

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